Lesson 1-7 • Instantaneous Rates of Change

Notes (filled in)

Learning goals

- Define the derivative at a point as the limit of a difference quotient.
- Compute derivatives of simple functions from the definition and estimate them from tables and graphs.
- Write the equation of a tangent line, and recognize where a function is not differentiable.

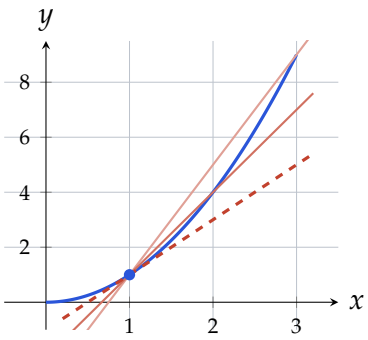
Vocabulary. instantaneous rate of change, derivative, $f'(a)$, tangent line, differentiable, corner, cusp, vertical tangent.

1. From average to instantaneous

Definition of the derivative

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h} \quad \text{or} \quad f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}.$$

$f'(a)$ is the instantaneous rate of change of f at a and the slope of the tangent line at $(a, f(a))$.



Secant lines from $(1,1)$ on $y = x^2$ approach the tangent line of slope 2.

Example 1 • From the definition

Find $f'(x)$ for $f(x) = x^2$, $g(x) = x^3$ and $k(x) = \frac{1}{x}$.

$\frac{(x+h)^2-x^2}{h} = 2x + h \to 2x$: $f'(x) = 2x$.
$\frac{(x+h)^3-x^3}{h} = 3x^2 + 3xh + h^2 \to 3x^2$: $g'(x) = 3x^2$.
$\frac{\frac{1}{x+h}-\frac{1}{x}}{h} = \frac{-1}{x(x+h)} \to -\frac{1}{x^2}$: $k'(x) = -\frac{1}{x^2}$.

Example 2 • Tangent line

Write the equation of the tangent line to $y = x^3$ at $x = 2$.

Point $(2, 8)$; slope $g'(2) = 3(4) = 12$. $y - 8 = 12(x - 2)$.

Example 3 • Estimating from a table

t (s)	0	2	4	6	8
$s(t)$ (m)	0	9	20	35	52

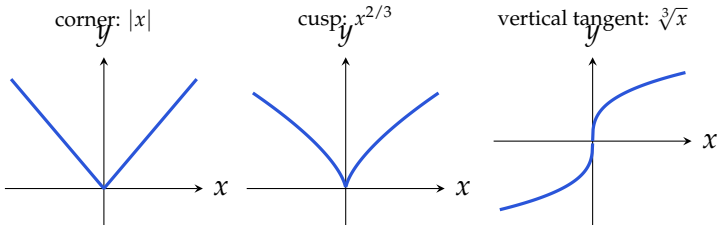
Estimate $s'(4)$ and interpret it.

Use the closest values on both sides: $\frac{s(6)-s(2)}{6-2} = \frac{35-9}{4} = 6.5$. At $t = 4$ s the object moves at about 6.5 m/s.

2. Differentiability

Key idea

If f is differentiable at a, then f is __continuous__ at a. The converse is __false__.
f is **not** differentiable where its graph has a __corner__, a __cusp__, a __vertical tangent__, or a __discontinuity__.



Example 4 • A piecewise function

Is $f(x) = \begin{cases} x^2, & x \le 1 \\ 2x - 1, & x > 1 \end{cases}$ differentiable at $x = 1$?

Continuous: both pieces give 1 at $x = 1$. Slopes: the left piece x^2 has slope $2(1) = 2$; the right piece has slope 2. The slopes match, so f is differentiable at $x = 1$ with $f'(1) = 2$.

Check your understanding

1. $\lim_{h\to 0} \frac{(3+h)^2-9}{h}$ is $f'(3)$ for $f(x) =$ __x^2__, and equals __6__.
2. $f(x) = |x - 2|$ is continuous at 2 but __not differentiable__ there.